

Linear Discriminant Analysis (LDA) – MCQs

1. What is the primary objective of Linear Discriminant Analysis (LDA)?
 - A) Minimize reconstruction error
 - B) Maximize class separability
 - C) Minimize classification accuracy
 - D) Increase dimensionality
2. LDA is primarily used for:
 - A) Clustering
 - B) Regression
 - C) Dimensionality Reduction and Classification
 - D) Data Augmentation
3. Which matrices are central to the formulation of LDA?
 - A) Covariance and Correlation Matrices
 - B) Within-Class Scatter and Between-Class Scatter Matrices
 - C) Adjacency and Incidence Matrices
 - D) Kernel and Gram Matrices
4. The within-class scatter matrix represents:
 - A) Separation between class means
 - B) Variance of samples within each class
 - C) Distance between clusters
 - D) Prediction error
5. The between-class scatter matrix measures:
 - A) Variability within classes
 - B) Separation among class means
 - C) Noise variance
 - D) Reconstruction loss
6. LDA seeks projection directions that:
 - A) Minimize both within-class and between-class variance
 - B) Maximize within-class variance
 - C) Maximize between-class variance while minimizing within-class variance

- D) Ignore class labels
7. For a dataset with C classes, the maximum number of LDA components is:
- A) C
 - B) $C + 1$
 - C) $C - 1$
 - D) Equal to the number of features
8. Which of the following is an assumption of classical LDA?
- A) Classes have identical covariance matrices
 - B) Classes have different covariance matrices
 - C) Features are independent and binary
 - D) Data must be non-linear
9. LDA is a supervised technique because:
- A) It uses unlabeled data
 - B) It requires class labels
 - C) It uses reinforcement signals
 - D) It performs clustering
10. In LDA, eigenvectors are obtained from:
- A) Covariance Matrix
 - B) Correlation Matrix
 - C) $S_W^{-1} S_B$
 - D) Gram Matrix
11. Compared to PCA, LDA:
- A) Does not reduce dimensions
 - B) Ignores class labels
 - C) Uses class information
 - D) Requires more dimensions
12. Which criterion is maximized in Fisher's Linear Discriminant?
- A) Reconstruction Error
 - B) Variance Ratio
 - C) Between-Class Scatter to Within-Class Scatter Ratio
 - D) Mean Squared Error
13. LDA assumes data from each class follows:

- A) Uniform Distribution
- B) Poisson Distribution
- C) Gaussian Distribution
- D) Exponential Distribution

14. If there are two classes, the maximum number of discriminant components is:

- A) 0
- B) 1
- C) 2
- D) Number of features

15. Which statement best differentiates PCA and LDA?

- A) PCA is supervised; LDA is unsupervised
- B) PCA maximizes variance; LDA maximizes class separability
- C) PCA requires labels; LDA does not
- D) Both optimize the same objective

16. What happens if class labels are randomly shuffled before applying LDA?

- A) Performance generally degrades
- B) Performance improves
- C) No effect
- D) Number of features increases

17. Which matrix must generally be invertible in standard LDA?

- A) Between-Class Scatter Matrix
- B) Within-Class Scatter Matrix
- C) Correlation Matrix
- D) Distance Matrix

18. LDA projects data onto:

- A) Directions of maximum variance
- B) Random directions
- C) Directions maximizing discrimination among classes
- D) Kernel feature space only

19. In pattern recognition, LDA is often applied:

- A) Before classification
- B) After classification

- C) Only during testing
- D) Only for clustering

20. Which of the following can be a limitation of LDA?

- A) Requires labeled data
- B) Assumes linear boundaries
- C) Sensitive to covariance estimation
- D) All of the above

21. LDA can be viewed as:

- A) Feature Extraction Technique
- B) Data Compression Technique
- C) Classification Technique
- D) All of the above

22. The rank of the between-class scatter matrix is at most:

- A) Number of Features
- B) Number of Samples
- C) Number of Classes Minus One
- D) Twice the Number of Classes

23. Which application commonly uses LDA?

- A) Face Recognition
- B) Speech Recognition
- C) Medical Diagnosis
- D) All of the above

24. In Fisher's criterion, a good projection should:

- A) Bring class means closer together
- B) Increase overlap among classes
- C) Separate class means and reduce class spread
- D) Ignore variance

25. LDA is most effective when:

- A) Classes are well separated
- B) Data is completely random
- C) No labels are available
- D) Features are constant

26. Which of the following affects LDA performance?
- A) Class imbalance
 - B) Outliers
 - C) Small sample size
 - D) All of the above
27. The generalized eigenvalue problem in LDA is solved to obtain:
- A) Cluster Centroids
 - B) Principal Components
 - C) Discriminant Directions
 - D) Decision Trees
28. LDA is likely to perform poorly when:
- A) Classes are linearly separable
 - B) Covariance assumptions are heavily violated
 - C) Labels are accurate
 - D) Features are informative
29. Which quantity is minimized in the denominator of Fisher's criterion?
- A) Between-Class Scatter
 - B) Within-Class Scatter
 - C) Total Variance
 - D) Correlation
30. In multiclass classification, LDA creates:
- A) One discriminant axis only
 - B) Multiple discriminant axes up to $C - 1$
 - C) One axis per feature
 - D) No projection axes
31. LDA is particularly useful when:
- A) Number of features is very small
 - B) Feature reduction is needed before classification
 - C) Data contains no labels
 - D) Only clustering is required
32. Which assumption is shared by many LDA formulations?
- A) Equal class priors

- B) Equal covariance matrices across classes
- C) Independent samples only
- D) Nonlinear decision surfaces

33. The output of LDA is typically:

- A) Cluster Assignments
- B) Reduced-Dimensional Feature Vectors
- C) Regression Coefficients
- D) Association Rules

34. Why is LDA often preferred over PCA for classification tasks?

- A) It ignores variance
- B) It uses label information
- C) It requires fewer samples
- D) It is unsupervised

35. If there are 5 classes in a dataset, the maximum number of discriminant vectors produced by LDA is:

- A) 3
- B) 4
- C) 5
- D) 6

36. Which type of learning does LDA belong to?

- A) Supervised Learning
- B) Unsupervised Learning
- C) Reinforcement Learning
- D) Self-Supervised Learning

37. What role do class means play in LDA?

- A) No role
- B) Used to compute between-class scatter
- C) Used only for normalization
- D) Used only during testing

38. Which of the following is true about LDA decision boundaries?

- A) Always nonlinear
- B) Typically linear
- C) Always quadratic

- D) Always circular

39. LDA can suffer from the "small sample size" problem when:

- A) Number of samples is much smaller than number of features
- B) Number of features is zero
- C) Classes are balanced
- D) Data is normalized

40. Which statement about LDA is correct?

- A) It ignores class membership
- B) It maximizes variance without labels
- C) It seeks projections that improve class discrimination
- D) It is purely a clustering algorithm